

Precise Definitions and Counterexamples

Answer Key

High-school geometry

Quick Answers

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|--|---|
| 1. — | 5. (b) $x = 20$, — |
| 2. 5 | 6. — |
| 3. — | 7. (a) $x = 5$, (b) the length = 11 cm |
| 4. (a) $x = 8$, (b) the measure of the whole angle = 84 degrees | 8. — |
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What is verified

3 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 3, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry

HSG-CO.A – *problems 1, 2, 5, 7, 8*

HSG-CO.C – *problems 3, 4, 6*

Difficulty 1–5 of 5 across 13 tagged checks.

1. Three attempted definitions of the midpoint of $\overline{AB}$. Which is precise, and what do the others leave out?

Attempt B is the precise definition. It names a testable location (a point *of* the segment) and a testable condition (the two pieces are congruent).

Attempt A leaves out “on the segment”. Every point of the perpendicular bisector of $\overline{AB}$ is the same distance from A and from B , and all but one of them lie off the segment — so the words are satisfied by infinitely many non-midpoints.

Attempt C uses “look equal”, which is not a condition anyone can check. Two pieces that look equal in a sketch may differ in length; congruence is the measurable version of the same idea.

2. M is the midpoint of $\overline{AB}$, $AM = 3x - 4$, $MB = x + 6$. Find x .

Solution: the definition of midpoint says $\overline{AM} \cong \overline{MB}$, which means the two lengths are equal — that equation is the whole content of the definition.

$$3x - 4 = x + 6$$

$$2x - 4 = 6$$

$$2x = 10$$

So $x = 5$. Check: $AM = 3(5) - 4 = 11$ and $MB = 5 + 6 = 11$.

$x = 5$

3. Counterexample to “if two angle measures add to 180° , the angles form a linear pair”.

Model answer: draw a 100° angle in one corner of the page and an 80° angle in the opposite corner. Their measures add to 180° , so the “if” part holds. They are not a linear pair: a linear pair must be *adjacent* (sharing a vertex and a side) and the two non-shared sides must be *opposite rays*. These two angles share nothing at all, so both conditions fail.

Grading: full credit needs (i) two measures that really add to 180° , and (ii) a reason naming the adjacency or opposite-ray condition. Two right angles drawn apart, or any supplementary pair drawn separately, is equally correct. Saying only “they are not next to each other” without naming the definition’s condition is partial credit.

4. $\overrightarrow{BD}$ bisects $\angle ABC$; $m\angle ABD = 5x + 2$, $m\angle DBC = 7x - 14$. (a) Find x . (b) Find $m\angle ABC$.

(a) An angle bisector divides the angle into two congruent angles, so the two measures are equal:

$$5x + 2 = 7x - 14$$

$$16 = 2x$$

So $x = 8$.

$x = 8$

(b) Each half measures $5(8) + 2 = 42$ degrees, and $7(8) - 14 = 42$ agrees. The whole angle is twice one half: $2(42)$.

$m\angle ABC = 84^\circ$

5. “Perpendicular lines are two lines that intersect.” (a) Repair the definition. (b) One of the four angles is $(4x + 10)^\circ$; find x .

(a) Model answer: any two lines that cross at, say, 30° fit the words exactly and are not perpendicular — intersecting is necessary but not sufficient. The missing condition is the angle: *perpendicular lines are two lines that intersect to form a right angle*, that is, an angle of 90° . (Grading: the repair must name a right angle or 90° . “They make a square corner” with no angle condition is partial credit.)

(b) By the repaired definition the marked angle is a right angle, so its measure is 90:

$$4x + 10 = 90$$

$$4x = 80$$

So $x = 20$.

$x = 20$

6. Counterexamples. (a) “Congruent angles are vertical angles.” (b) “A point the same distance from A and B is the midpoint of $\overline{AB}$.”

(a) **Model answer:** the two base angles of an isosceles triangle are congruent, and they are not vertical angles — vertical angles are the opposite pair formed by *two intersecting lines*, and a triangle’s base angles are not such a pair. Two separately drawn 40° angles work just as well.

(b) **Model answer:** take $\overline{AB}$ of length 10 and a point P two units directly above its midpoint. Then $PA = PB$, so the “if” part holds, but P is not on $\overline{AB}$ at all, so it cannot be the midpoint. Every point of the perpendicular bisector except the midpoint itself is a counterexample.

Grading: each part needs a case that satisfies the hypothesis *and* an explicit reason for failing the conclusion. In (b) the reason must be that the point is off the segment.

7. $\overline{PQ} \cong \overline{RS}$, $PQ = 2x + 1$, $RS = 5x - 14$. (a) Find x . (b) Find PQ .

(a) Congruent segments are segments of equal length, so

$$2x + 1 = 5x - 14$$

$$15 = 3x$$

So $x = 5$.

(b) Substitute into either expression: $PQ = 2(5) + 1$, and $RS = 5(5) - 14 = 11$ agrees, which is the check that the congruence really holds.

$x = 5$

$PQ = 11 \text{ cm}$

8. “The midpoint of a segment is a point on the segment that divides it into two segments.”

(a) Counterexample. (b) Rewrite it.

(a) **Model answer:** let $\overline{AB}$ have length 12 and let P sit 3 units from A . Then P is on the segment and it does divide $\overline{AB}$ into two segments, of lengths 3 and 9 — every word of the definition is satisfied. But $3 \neq 9$, so P is not the midpoint. Any interior point other than the midpoint works.

(b) **Model answer:** “The midpoint of $\overline{AB}$ is the point M on $\overline{AB}$, between A and B , such that $\overline{AM} \cong \overline{MB}$.” The three conditions are: on the segment; between the endpoints; the two pieces congruent. The counterexample in (a) fails the third.

Grading: the rewrite must carry the congruence condition — that is the one the counterexample exposed. Full credit also states the on-the-segment and betweenness conditions.